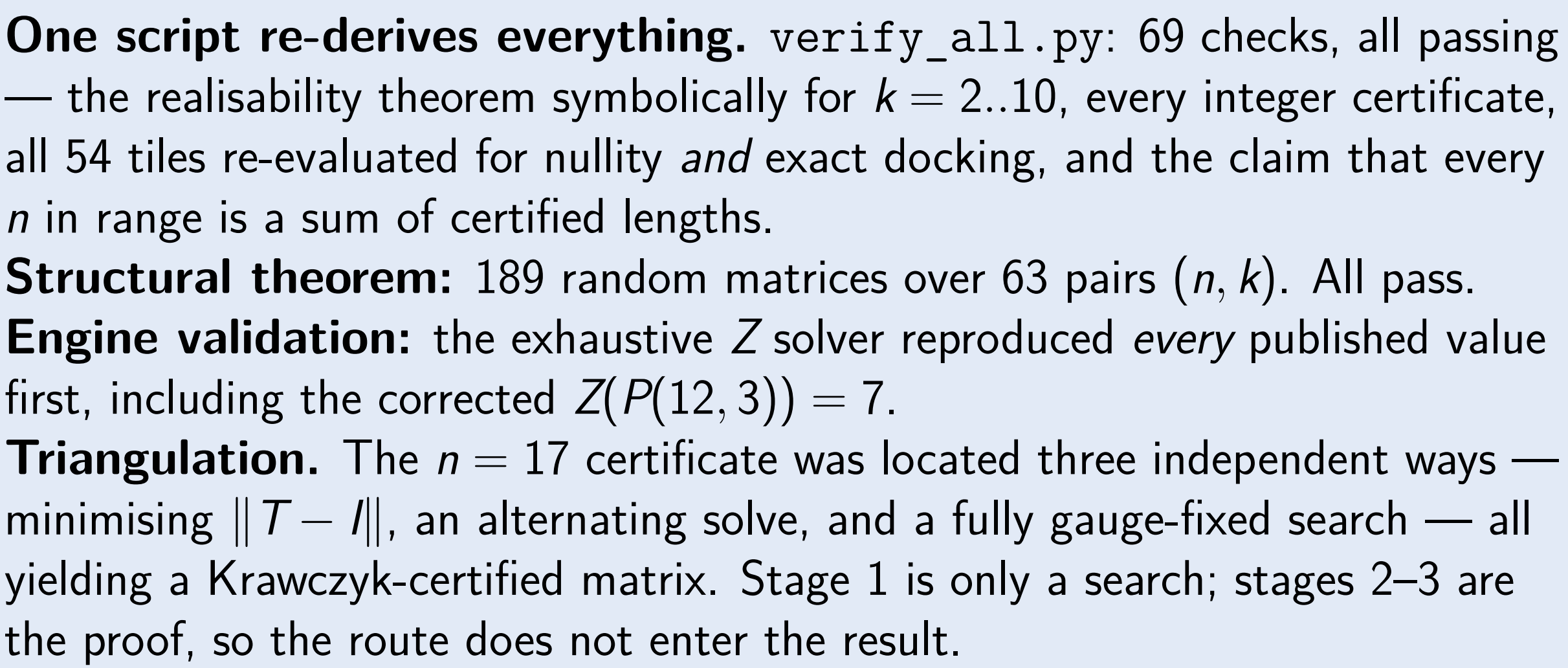


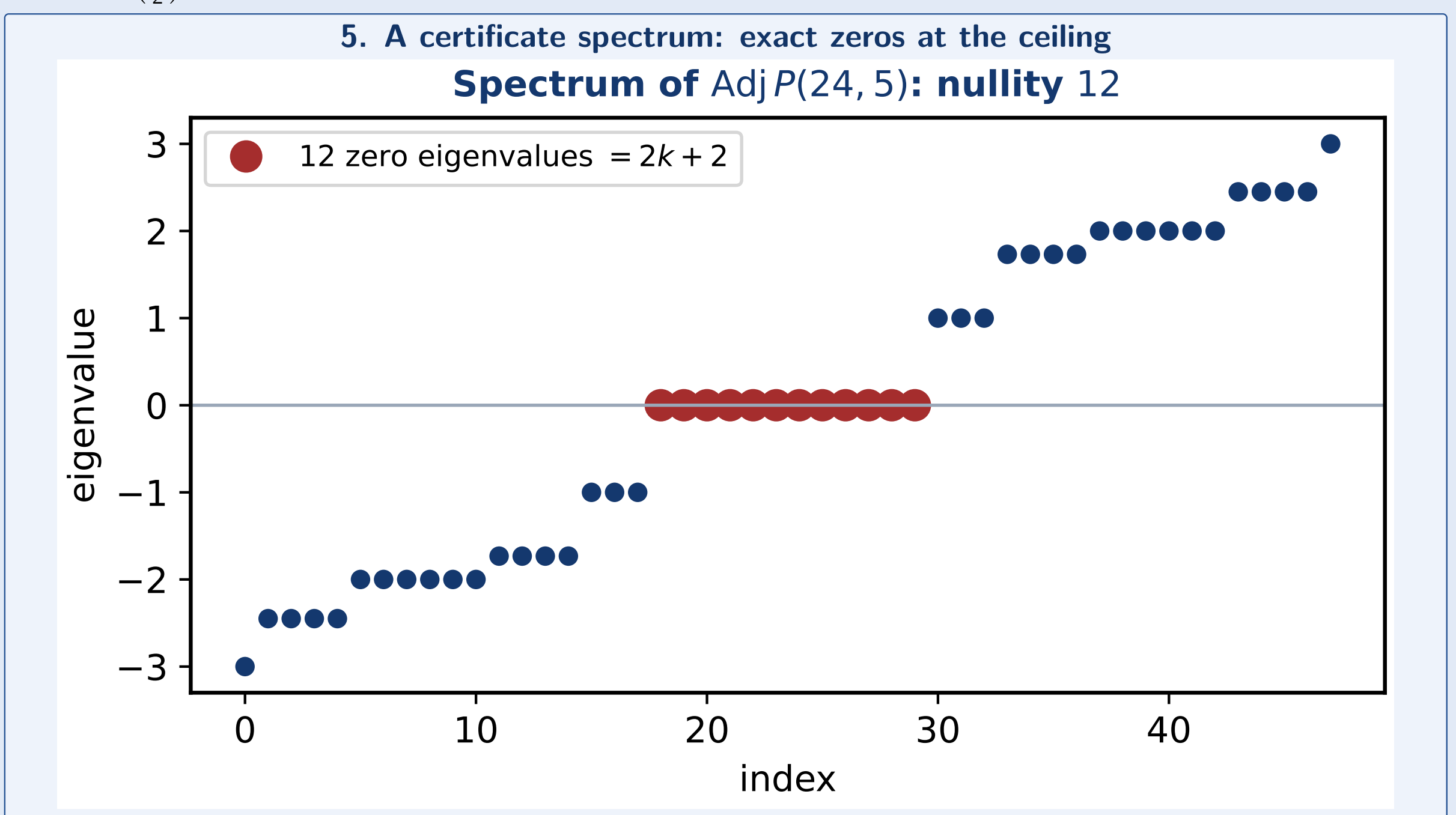
VERIFICATION



Why a failed search proves nothing here. *Lemma.* Every symmetric matrix on the pattern is gauge-equivalent to one in the slice we search. So a failure inside it measures effort, not impossibility — and this bit: $P(18, 3)$ and the tile $\ell = 21$ were both reported as failures and both succeeded once the restart budget was raised. *Working rule: quote the residual and the budget, never just “failed”.*

THE CERTIFICATIONS

Each tile is an independent Krawczyk contraction: its own unknown count, its own $\|Y\|_\infty$, its own residual. Nothing is copied between them. At $k = 4$, $\ell = 61$: 1425 unknowns, 1220 equations of which 1175 are structural and $45 = \binom{10}{2}$ dependent, high-precision residual 4.7×10^{-61} .



CONCLUSIONS

1. The method has an **exact ceiling** of $2k + 2$, coinciding with the best known upper bound: either it settles Z completely or nothing does.
2. **Proved:** $Z(P(n, 2))$ and $Z(P(n, 3))$ **determined for every n** ; $Z(P(n, 4)) = 10$ for every $n \geq 30$. **No divisibility conditions.** The $k = 3$ sequence is 6, 6, 6, 8, 7, 7, 8, 8, ... — non-monotone, which is how the published proof went wrong.
3. **Proved:** the congruence conditions of the cyclotomic layer were *forced*, not a shortfall — one class, at most one prime. That converts the earlier results into a demarcation.
4. **Proved:** $c^2 = 0$ iff the symbol factors as $(s + a)(L_k + t)$; over $\mathbb{Q}$, Niven's theorem caps the degenerate symbols at 25 per k .
5. **Corrected (my own), among others:** the claim this construction “never exceeds 6”; a false unconditional cover bound ($Z(P(8, 2)) = 5$); the threshold $(2k + 2)^2/5$; $N(k) = O(k)$; and an eigenvalue search whose negative results I **withdrew** after it reported nullity 7 at $P(20, 3)$ where 8 is attained.

RESULTS & CERTIFICATION

Main result. Every value below is new or corrects a published one.

- $Z(P(n, 2))$ determined for **every** n , $Z(P(n, 3))$ determined for **every** n ,
- $Z(P(n, 3)) = 8$ exactly for $n \geq 13$, and 13 is optimal : $Z(P(11, 3)) = Z(P(12, 3)) = 7$,
- $Z(P(n, 4)) = M(P(n, 4)) = 10$ for **every** $n \geq 30$.

The $k = 3$ case is a **first determination**: the value published in 2020 is false, and the 2026 note that found the error left the stabilization threshold open. **We determine it: it is 13.**

FUTURE WORK

- Honest limits.** $k \geq 5$ is *not* settled: tiles exist at $k = 5$ from $l \approx 54$, but [54, 108] needs 54 of them and we have 19. Four values at $k = 4$ remain, $n \in \{24, 25, 28, 29\}$. The $k \geq 6$ congruence-class values are numerical, a weaker standard than the rest. The rank law is verified over a range, not proved.
- What we cannot diagnose.** We first searched $k = 5$ over $41 \leq l \leq 45$ — the parameter count’s bound — and found nothing, then found tiles starting near 54. The count is a lower bound, *not* a predictor of where tiles begin (21 vs 16 at $k = 3$; ≈ 54 vs 41 at $k = 5$), and we cannot explain the gap.
- Next.** Close [54, 108] for $k = 5$ (mechanical); prove the rank law; decide whether $M < Z$ strictly at $n = 10$, $k = 2$.

KEY REFERENCES